PROJECT

Aim

To analyze and visualize a retail sales dataset using Python, Power BI, and Tableau by creating bar charts, line charts, pie charts, histograms, and interactive dashboards to identify sales trends, category-wise performance, and profit distribution.

Description

Data visualization is the graphical representation of data using charts, graphs, and dashboards. In this experiment, a retail sales dataset containing order details, dates, products, categories, regions, quantities, sales, and profits is used.

Python libraries such as Pandas and Matplotlib are used to process the dataset and create different visualizations. Power BI and Tableau are used to build interactive dashboards with charts, tables, and filters for effective data analysis and decision-making.

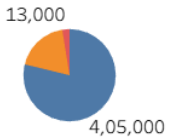
Observation

1. The bar chart helps compare sales across different product categories.
2. The line chart identifies monthly sales trends and fluctuations.
3. The pie chart displays the percentage contribution of each category to total sales.
4. The histogram shows the distribution and frequency of order sales values.
5. The scatter plot helps analyze the relationship between sales and profit.
6. Power BI and Tableau dashboards allow users to explore sales data interactively using filters and visual reports.

Result

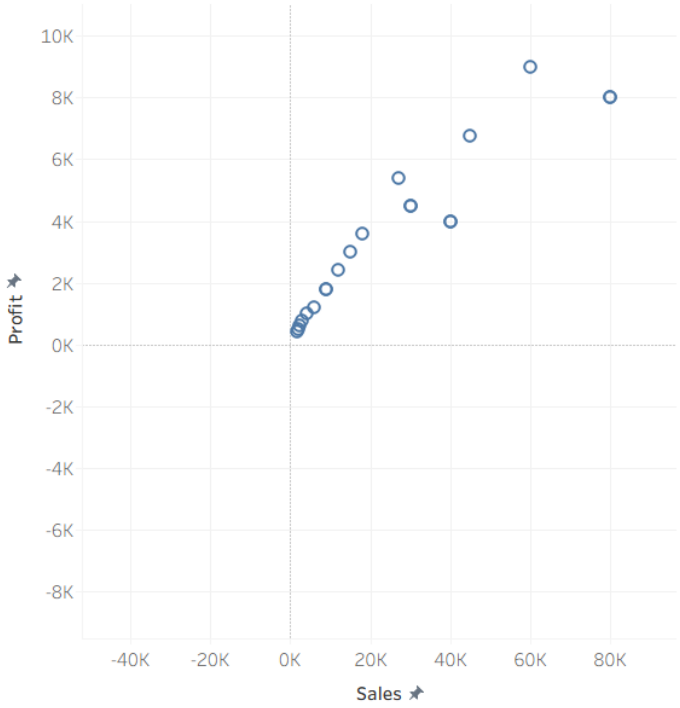
The retail sales dataset was successfully analyzed and visualized using Python, Power BI, and Tableau. The visualizations helped identify category-wise sales performance, monthly trends, order-value distribution, and profitability.

Category Sales Share

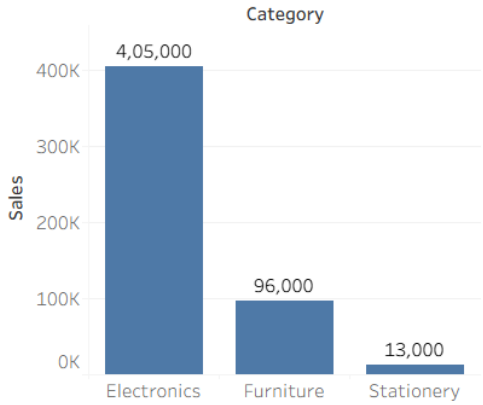


Category
- Electronics
- Furniture
- Stationery

Sales VS Profit



Category by Sales



Total Sales
514K

Total Profit
71K

Total Orders
20

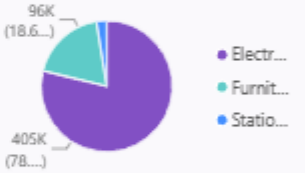
Total Sales
by Category



Total Sales
by Year, Quarter, Month and Day



Total Sales
by Category



Category	Year	Quarter	Month	Day	Product	Region	Sum of Pr
Electronics	2026	Qtr 1	January	5	Laptop	South	8
Electronics	2026	Qtr 1	January	8	Phone	North	6
Electronics	2026	Qtr 1	February	3	Laptop	West	4
Electronics	2026	Qtr 1	February	25	Phone	East	4
Electronics	2026	Qtr 1	March	11	Laptop	East	8
Electronics	2026	Qtr 2	April	4	Phone	South	9
Electronics	2026	Qtr 2	April	22	Laptop	North	4
Electronics	2026	Qtr 2	May	15	Phone	North	4
Total							**71**

Category

Electronics

Furniture

Stationery

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("retail_sales_dataset.csv")

print("First five rows:")
print(df.head())
print("\nDataset information:")
print(df.info())
print("\nMissing values:")
print(df.isnull().sum())

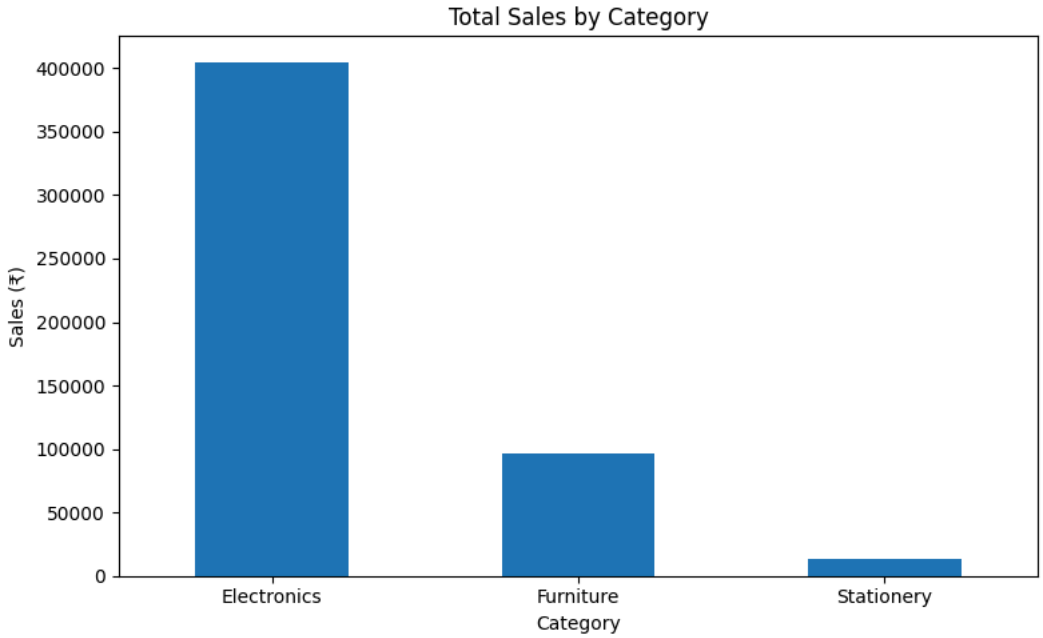
df["Order_Date"] = pd.to_datetime(df["Order_Date"])
df["Month"] = df["Order_Date"].dt.to_period("M").astype(str)

category_sales = df.groupby("Category")["Sales"].sum().sort_values(ascending=False)
plt.figure(figsize=(8, 5))
category_sales.plot(kind="bar")
plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales (₹)")
plt.xticks(rotation=0)
plt.tight_layout()
plt.savefig("1_bar_chart.png", dpi=150)
plt.show()
```

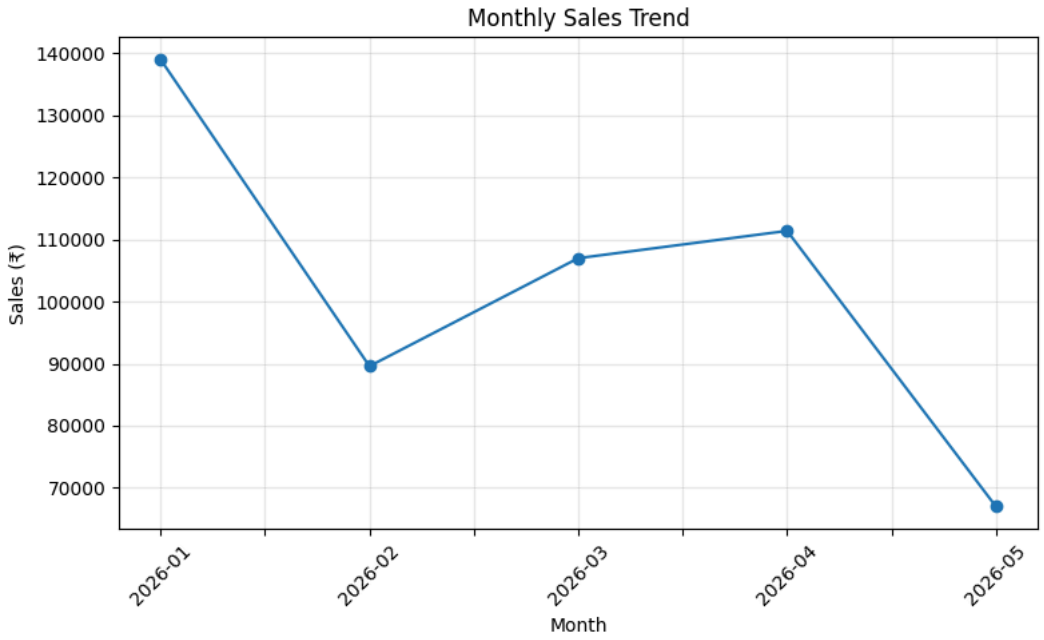
```
First five rows:
   Order_ID  Order_Date   Product     Category Region  Quantity  Sales  Profit
0      1001  2026-01-05    Laptop  Electronics  South         2  80000    8000
1      1002  2026-01-08     Phone  Electronics  North         3  45000    6750
2      1003  2026-01-12     Chair    Furniture   East         4  12000    2400
3      1004  2026-01-20  Notebook   Stationery   West        10   2000     500
4      1005  2026-02-03    Laptop  Electronics   West         1  40000    4000

Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Order_ID    20 non-null     int64
 1   Order_Date  20 non-null     object
 2   Product     20 non-null     object
 3   Category    20 non-null     object
 4   Region      20 non-null     object
 5   Quantity    20 non-null     int64
 6   Sales       20 non-null     int64
 7   Profit      20 non-null     int64
dtypes: int64(4), object(4)
memory usage: 1.4+ KB
None

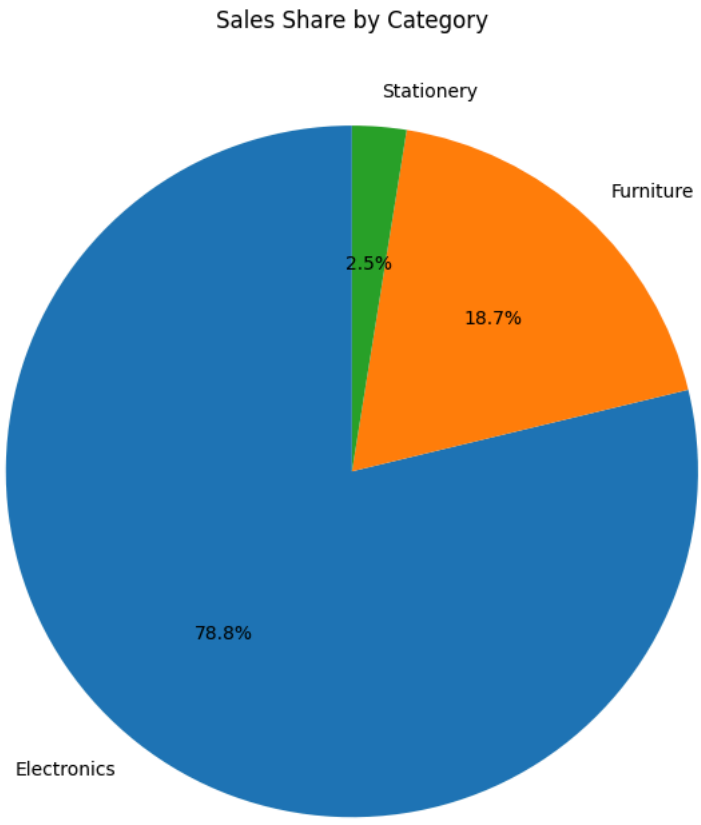
Missing values:
Order_ID      0
Order_Date    0
Product       0
Category      0
Region        0
Quantity      0
Sales         0
Profit        0
dtype: int64
```



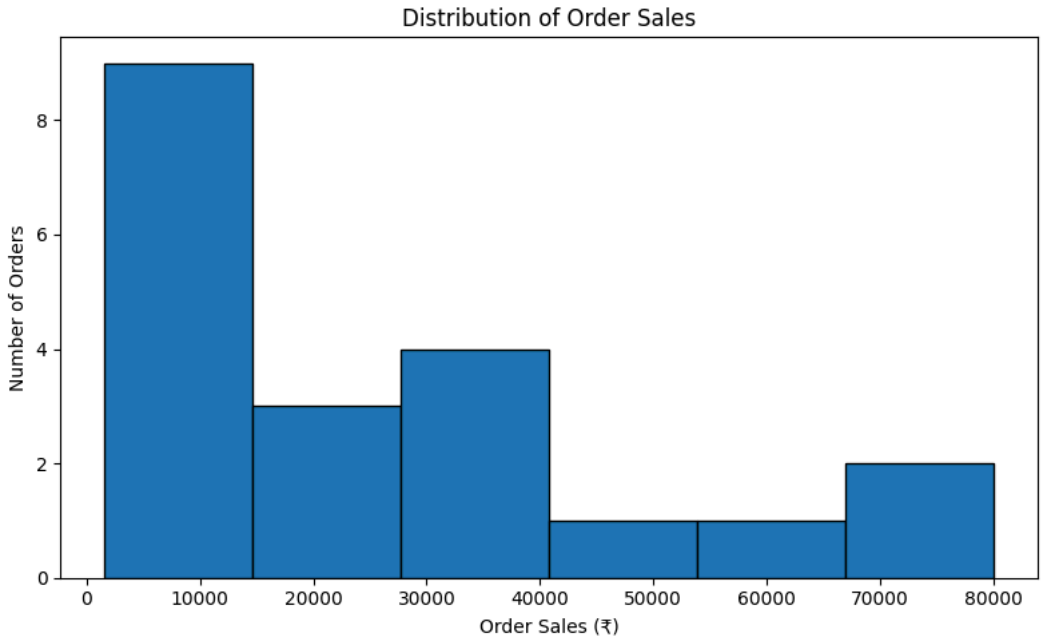

```
monthly_sales = df.groupby("Month")["Sales"].sum()
plt.figure(figsize=(8, 5))
monthly_sales.plot(kind="line", marker="o")
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales (₹)")
plt.xticks(rotation=45)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig("2_line_chart.png", dpi=150)
plt.show()
```




```
plt.figure(figsize=(7, 7))
category_sales.plot(kind="pie", autopct="%1.1f%%", startangle=90)
plt.title("Sales Share by Category")
plt.ylabel("")
plt.tight_layout()
plt.savefig("3_pie_chart.png", dpi=150)
plt.show()
```




```
plt.figure(figsize=(8, 5))
plt.hist(df["Sales"], bins=6, edgecolor="black")
plt.title("Distribution of Order Sales")
plt.xlabel("Order Sales (₹)")
plt.ylabel("Number of Orders")
plt.tight_layout()
plt.savefig("4_histogram.png", dpi=150)
plt.show()
```




```
plt.figure(figsize=(8, 5))
plt.scatter(df["Sales"], df["Profit"])
plt.title("Sales vs Profit")
plt.xlabel("Sales (₹)")
plt.ylabel("Profit (₹)")
plt.tight_layout()
plt.savefig("5_scatter_plot.png", dpi=150)
plt.show()
```

